PYTHON PROGRAMMING
10 PROBLEM STATEMENTS

Algorithms • Sample Inputs • Sample Outputs • Flowcharts

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Note: The examples use illustrative policy values. Replace rates, thresholds, and rules with those specified by your instructor.

1. University Scholarship Allocation

Objective: Evaluate applicants using marks, subject minimums, attendance, and family income; rank eligible applicants and resolve ties by aggregate marks, then attendance.

Input Required

• Number of applicants; each applicant's name, aggregate percentage, lowest subject mark, attendance percentage, annual family income.

• Minimum aggregate, minimum subject mark, minimum attendance, maximum income, and number of scholarships.

Algorithm

1. Start.

2. Read policy limits and applicant details.

3. For each applicant, check all four eligibility conditions.

4. Store eligible applicants; reject others with a reason.

5. Sort eligible applicants by aggregate marks descending, then attendance descending; if still tied, use name alphabetically.

6. Display eligible ranking and scholarship recipients up to the available count.

7. Stop.

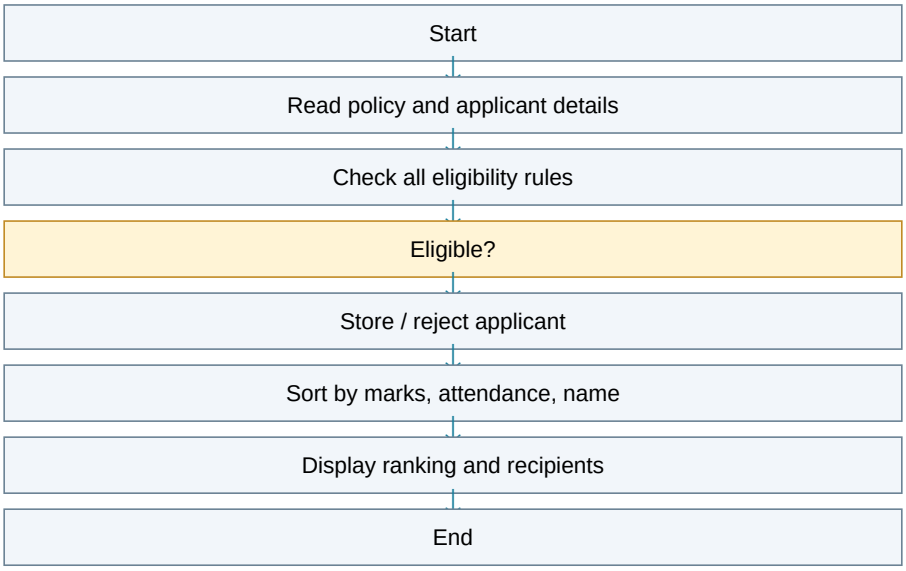
Sample Input

```
Minimum aggregate = 80; minimum subject = 70; attendance = 85%; maximum income = ■250000; scholarships =
2
Asha, 91, 75, 92, ■180000
Balu, 88, 72, 90, ■220000
Charan, 88, 72, 90, ■200000
Divya, 79, 80, 95, ■150000
```

Sample Output

```
Eligible ranking:
1. Asha – 91%, attendance 92%
2. Charan – 88%, attendance 90%
3. Balu – 88%, attendance 90%
Selected: Asha, Charan
Divya – Not eligible: aggregate below 80%.
```

Flowchart



2. Progressive Electricity Billing

Objective: Calculate a bill using progressive consumption slabs, fixed charge, surcharge, and rebate; compare monthly bills and flag unusual increases.

Input Required

• Units consumed for each month.

• Slab rates and boundaries, fixed charge, surcharge percentage, rebate rule, and unusual-increase threshold.

Algorithm

1. Start.

2. Read monthly units and billing rules.

3. Calculate energy charge by applying each rate only to units in that slab.

4. Add fixed charge; calculate surcharge and eligible rebate.

5. Calculate final bill and store it.

6. Compare each bill with the previous month; calculate percentage increase.

7. Flag increases above the given threshold.

8. Display monthly bills and alerts; stop.

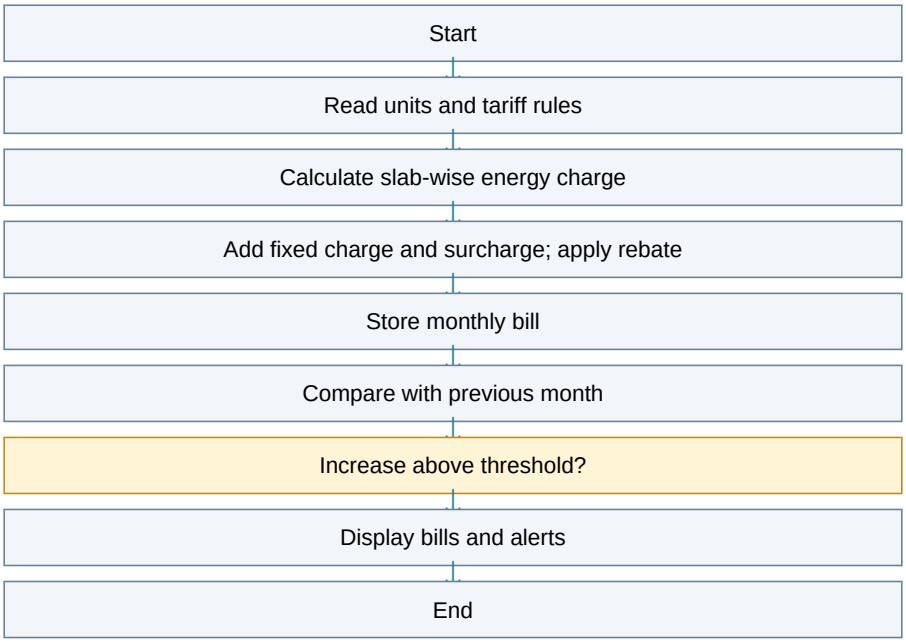
Sample Input

```
Tariff: first 100 units ■2/unit; next 100 ■3/unit; above 200 ■5/unit; fixed charge ■50; surcharge 5%;
rebate ■20 if units ≤ 100; flag increase > 30%.
Units: April 80, May 120, June 250
```

Sample Output

```
April: energy ■160, fixed ■50, surcharge ■10.50, rebate ■20; bill = ■200.50
May: energy ■260, fixed ■50, surcharge ■15.50; bill = ■325.50
June: energy ■750, fixed ■50, surcharge ■40; bill = ■840.00
May vs April: +62.34% — ALERT
June vs May: +158.06% — ALERT
```

Flowchart



3. Hospital Billing and Patient Classification

Objective: Add consultation, diagnostic, and treatment charges; apply a concession when eligible; validate payment and classify the account.

Input Required

• Consultation, diagnostic, and treatment charges; concession eligibility and percentage; amount paid; classification thresholds.

Algorithm

1. Start.
2. Read charges, concession conditions, and payment amount.
3. Validate that charges and payment are non-negative.
4. Add consultation, diagnostic, and treatment charges.
5. If eligible, subtract the allowed concession.
6. Calculate balance = final bill – amount paid; validate overpayment using the policy.
7. Classify as Paid, Partially Paid, or Unpaid.
8. Display itemized bill and classification; stop.

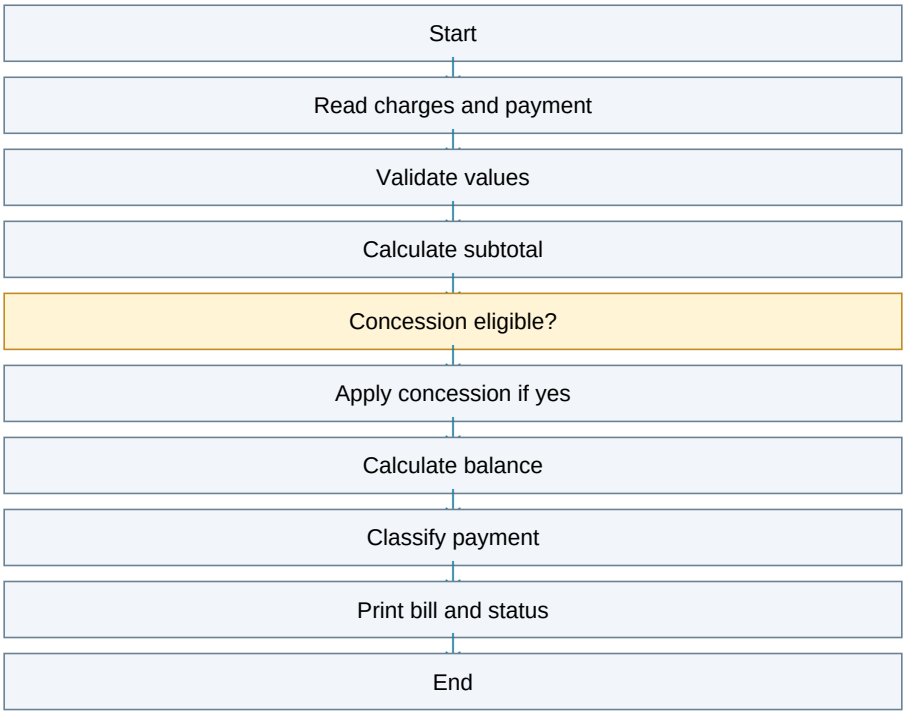
Sample Input

```
Consultation ■500; diagnostics ■1500; treatment ■4000; eligible concession 10%; paid ■3000. Concession
applies to subtotal.
```

Sample Output

```
Subtotal = ■6000.00
Concession (10%) = ■600.00
Final bill = ■5400.00
Paid = ■3000.00
Balance due = ■2400.00
Classification: Partially Paid
```

Flowchart



4. Employee Payroll and Deduction Analysis

Objective: Calculate gross pay from basic salary, allowances, overtime, and incentives; apply progressive deductions and print a salary statement.

Input Required

• Basic salary, allowances, overtime hours and hourly rate, incentive, deduction slabs/rates, and fixed deductions.

Algorithm

1. Start.
2. Read employee and pay details.
3. Calculate overtime pay = hours × hourly rate.
4. Calculate gross salary = basic + allowances + overtime + incentives.
5. Apply supplied progressive deduction brackets.
6. Add fixed deductions, if any.
7. Calculate net salary = gross – total deductions.
8. Print salary statement; stop.

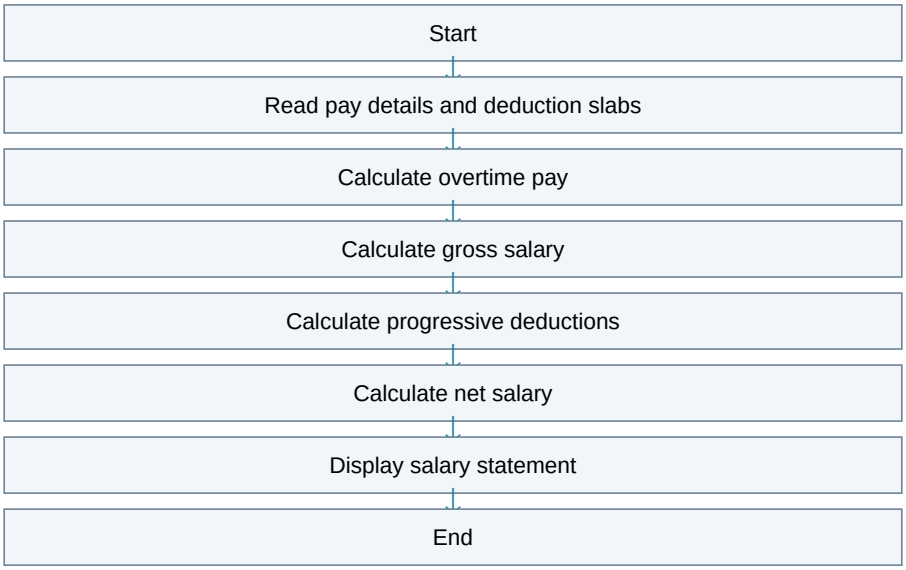
Sample Input

```
Basic ■25000; allowances ■5000; overtime 10 hours at ■200/hour; incentive ■2000. Deduction: 0% on first ■20000, 5% on next ■10000, 10% above ■30000.
```

Sample Output

```
Basic: ■25000.00
Allowances: ■5000.00
Overtime: ■2000.00
Incentive: ■2000.00
Gross salary: ■34000.00
Progressive deduction: ■1000.00
Net salary: ■33000.00
```

Flowchart



5. Multi-Plan Telecommunications Cost Optimization

Objective: Compare mobile plans using expected usage, validity, and budget; calculate extra-use charges and choose the lowest-cost eligible plan.

Input Required

• Expected data/call/SMS usage, required validity, budget, and each plan's price, included allowances, validity, and excess rates.

Algorithm

1. Start.
2. Read requirements and plan details.
3. Check validity and mandatory requirements for each plan.
4. Calculate excess charges for usage beyond allowances.
5. Calculate total cost = plan price + excess charges.
6. Mark plans exceeding budget or failing requirements as ineligible.
7. Choose eligible plan with lowest total cost.
8. Display comparison and selected plan; stop.

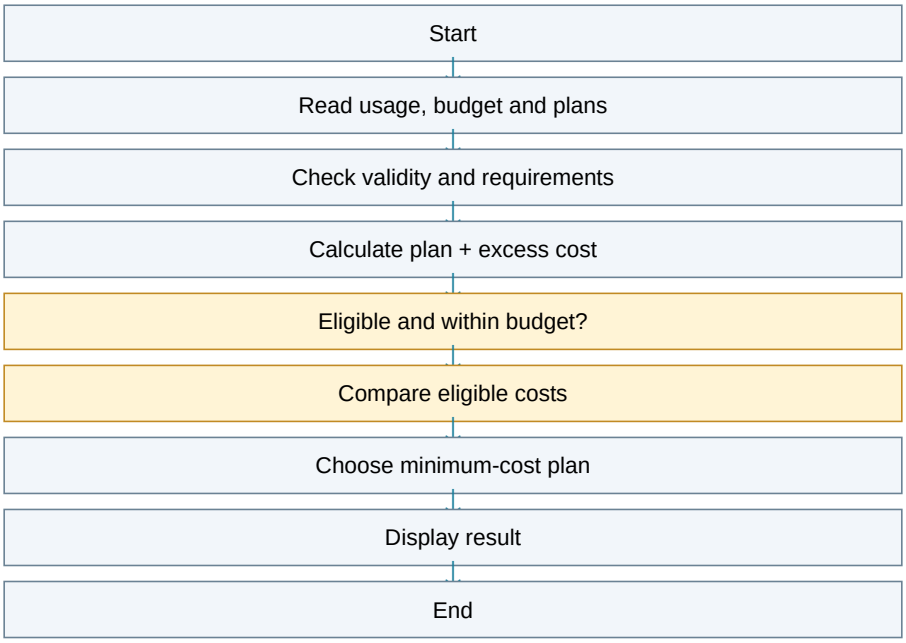
Sample Input

```
Usage: 12 GB data; 300 call minutes; validity 28 days; budget ■400.
Plan A: ■299, 10 GB, 300 minutes, 28 days, extra data ■10/GB.
Plan B: ■349, 15 GB, 300 minutes, 28 days.
Plan C: ■249, 8 GB, 250 minutes, 24 days.
```

Sample Output

```
Plan A: ■299 + 2 GB × ■10 = ■319; eligible
Plan B: ■349; eligible
Plan C: ineligible (validity and allowances insufficient)
Selected plan: Plan A — ■319
```

Flowchart



6. Railway Fare and Reservation Calculator

Objective: Calculate passenger fares using distance, travel class, age concessions, reservation fees, and cancellation penalties.

Input Required

• Passenger age and count, distance, class rate per kilometre, age concession rules, reservation fee, and cancellation penalty.

Algorithm

1. Start.
2. Read journey and passenger details.
3. For each passenger, calculate base fare = distance × class rate.
4. Apply age concession if eligible.
5. Add reservation fee and other charges.
6. If cancelled, calculate penalty and refund according to policy.
7. Calculate total fare/refund.
8. Display itemized results; stop.

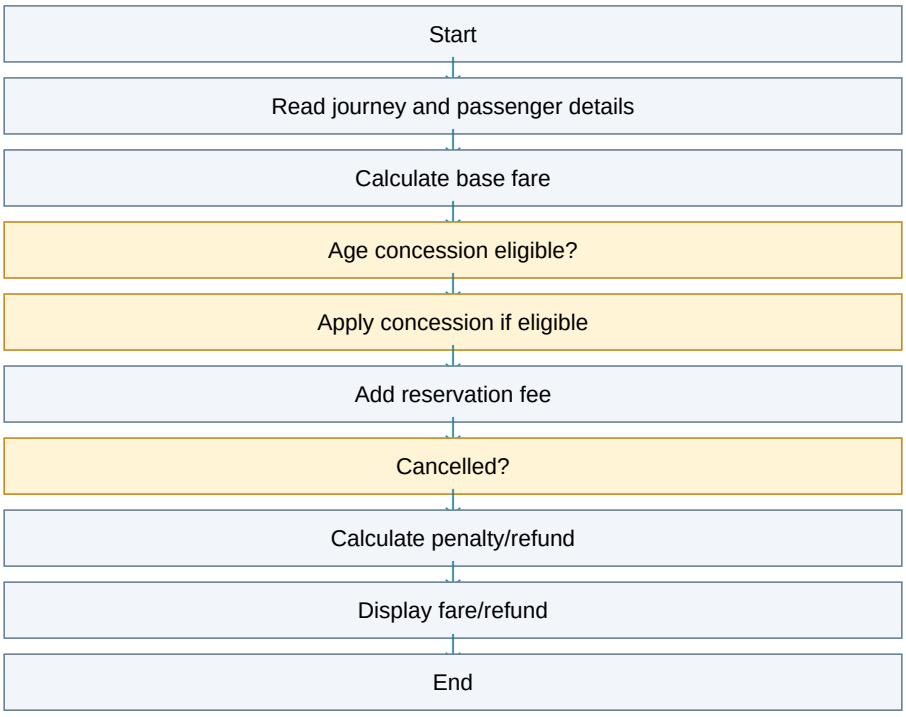
Sample Input

```
Distance 250 km; class rate ■2/km; reservation ■40 per passenger; concession 40% for age 60+;
cancellation penalty ■100. Passengers aged 25 and 65. Concession applies to base fare.
```

Sample Output

```
Age 25: base ■500, concession ■0, reservation ■40, fare ■540
Age 65: base ■500, concession ■200, reservation ■40, fare ■340
Total fare = ■880
If age-65 passenger cancels: refund = ■340 – ■100 = ■240
```

Flowchart



7. Retail Billing with Multiple Discount Rules

Objective: Create an itemized invoice with product quantities, category taxes, threshold discounts, and a maximum discount limit.

Input Required

• Product name, category, unit price, quantity, category tax rate; discount thresholds/rates; maximum discount limit.

Algorithm

1. Start.
2. Read product details and billing rules.
3. For each item, calculate line amount = unit price × quantity.
4. Add line amounts and calculate category-specific tax.
5. Determine discount using the threshold rule.
6. Limit discount to maximum allowed concession.
7. Calculate final total = subtotal + tax – discount.
8. Print invoice and total; stop.

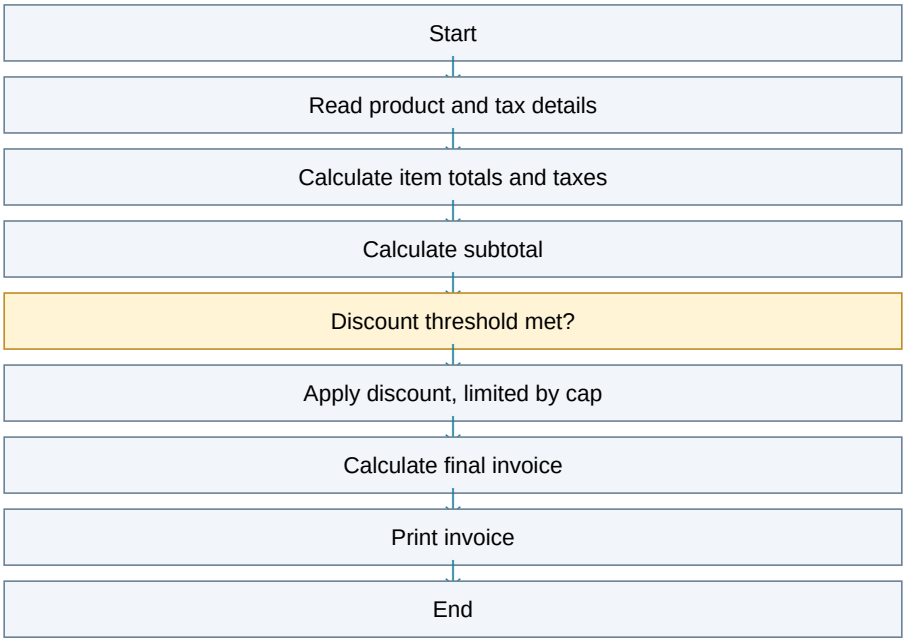
Sample Input

```
Rice: ■60 × 2, tax 0%; Soap: ■40 × 3, tax 5%; Oil: ■150 × 1, tax 12%. Discount 10% if subtotal ≥ ■400,
capped at ■50.
```

Sample Output

```
Rice: ■120.00; tax ■0.00
Soap: ■120.00; tax ■6.00
Oil: ■150.00; tax ■18.00
Subtotal = ■390.00
Tax = ■24.00
Discount = ■0.00 (below ■400 threshold)
Final invoice = ■414.00
```

Flowchart



8. Manufacturing Cost and Profitability Analysis

Objective: Calculate material, labour, wastage, and machine costs; determine selling revenue and whether the profit target is met.

Input Required

• Units produced, material cost per unit, wastage percentage, labour cost, machine cost, selling price per unit, and target profit.

Algorithm

1. Start.
2. Read production and cost details.
3. Calculate material cost and wastage cost.
4. Add labour and machine costs to find total production cost.
5. Calculate revenue = units sold × selling price.
6. Calculate profit = revenue – total cost.
7. Compare profit with target profit.
8. Display costs, revenue, profit, and target status; stop.

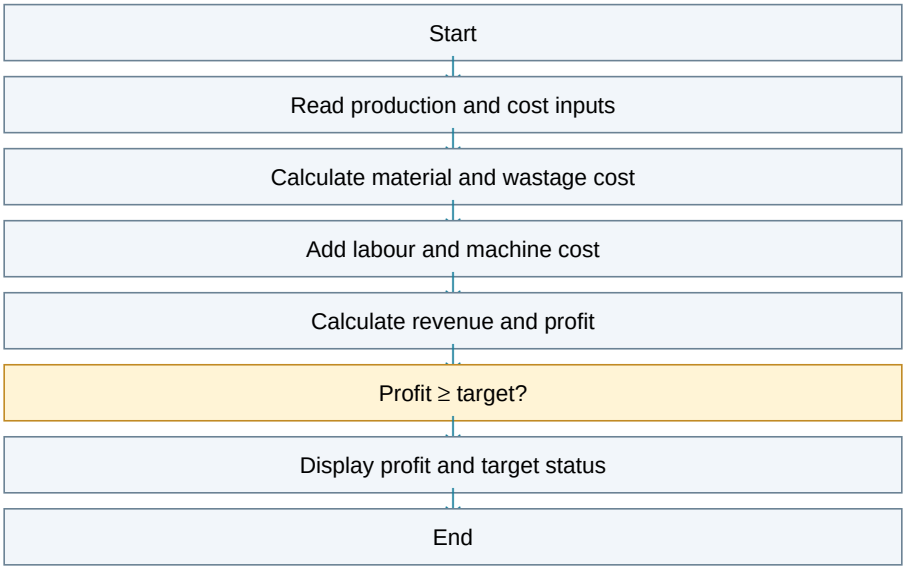
Sample Input

```
100 units; material ■80/unit; wastage 5% of material cost; labour ■3000; machine ■1500; selling price ■130/unit; target profit ■2500. All units sold.
```

Sample Output

```
Material = ■8000.00
Wastage = ■400.00
Labour = ■3000.00
Machine = ■1500.00
Total cost = ■12900.00
Revenue = ■13000.00
Profit = ■100.00
Target ■2500: NOT MET
```

Flowchart



9. Multi-Currency Travel Budget Analysis

Objective: Convert expenses using supplied exchange rates and fees, compare the total with a budget, and identify the largest expense category.

Input Required

• Expense categories and amounts in foreign currency; exchange rate to home currency; conversion fee percentage; total budget.

Algorithm

1. Start.
2. Read expenses, exchange rates, fees, and budget.
3. Convert each category to home currency.
4. Calculate conversion fee and adjusted cost.
5. Add adjusted costs to get total spending.
6. Compare total spending with budget.
7. Find the category with the largest adjusted cost.
8. Display converted expenses and budget comparison; stop.

Sample Input

```
Home currency INR; 1 USD = ■83; conversion fee 2%. Hotel $300, food $120, transport $80, activities
$100. Budget ■50000.
```

Sample Output

```
Hotel: $300 → ■24900 + ■498 fee = ■25398
Food: $120 → ■9960 + ■199.20 = ■10159.20
Transport: $80 → ■6640 + ■132.80 = ■6772.80
Activities: $100 → ■8300 + ■166 = ■8466
Total = ■50796.00
Over budget by ■796.00
Largest category: Hotel
```

Flowchart

10. Campus Event Financial Planning

Objective: Estimate event costs, calculate participant contribution and sponsorship shortfall, and find minimum registrations needed to break even.

Input Required

• Venue, equipment, catering per participant, transport, certificate per participant, sponsorship, registration fee, and expected participant count.

Algorithm

1. Start.

2. Read fixed costs, per-participant costs, sponsorship, fee, and attendance.

3. Calculate fixed cost = venue + equipment + transport.

4. Calculate variable cost per participant = catering + certificate.

5. Calculate total cost and revenue for expected participants.

6. Calculate surplus/shortfall = revenue – total cost.

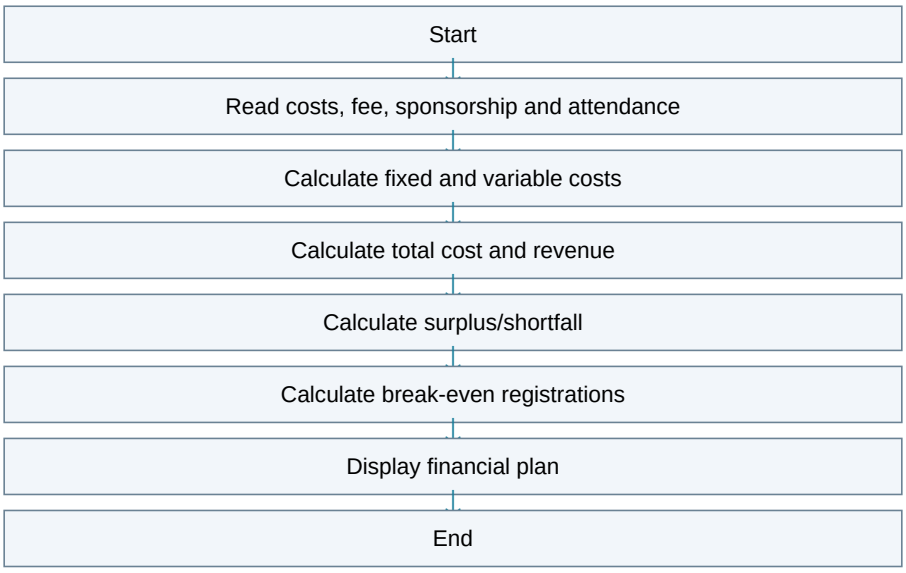
7. Calculate break-even registrations using (fixed costs – sponsorship) ÷ (fee – variable cost), rounded up; ensure fee exceeds variable cost.

8. Display costs, balance, and minimum registrations; stop.

Sample Input

```
Venue ■10000; equipment ■5000; transport ■3000; catering ■120/person; certificate ■20/person;
sponsorship ■8000; registration ■250/person; 100 participants.
```

Sample Output

```
Fixed cost = ■18000
Variable cost/person = ■140
Total cost for 100 = ■32000
Revenue = ■8000 sponsorship + ■25000 registration = ■33000
Surplus = ■1000
Break-even = ceil((■18000 – ■8000) / (■250 – ■140)) = 91 participants
```

Flowchart